

MATTHEW KURTIS BURNS

AI Security Engineer — Adversarial Evaluation × LLM Post-Training Research

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PROFILE

Cybersecurity researcher specialising in adversarial evaluation of autonomous AI agents. I built a 12-dimension evaluation framework for offensive-security LLMs, fine-tuned four Qwen3-14B variants (Baseline / SFT / DPO / GRPO), and quantified the safety–performance trade-off across 32 controlled cells on Hack The Box machines — published as the open-source *Beyond CTFs* framework. Looking to apply this combination of **LLM post-training, red-teaming and AI governance** to frontier-AI evaluation roles in Australia.

EDUCATION

Master of Cybersecurity — Cloud Security Specialisation · 2025 – 2026

The University of Sydney · Student ID 541 044 240

- **WAM ~76 (Distinction average)** — Distinctions in *Cybersecurity Capstone Project A & B, Applied Cryptography, and Cloud Security*
- Further Distinctions in Applied Cybersecurity, Professional Practice in IT, and Understanding IT Innovations

Bachelor of Information Technology (Cybersecurity) · Mar 2020 – Aug 2024

Southern Cross University · GPA > 5.0

CompTIA Security+ (ce) · Jul 2024

FEATURED RESEARCH

Beyond CTFs — A Real-World Evaluation Framework for Autonomous Pentesting Agents · Mar 2026 – Jun 2026

Master's Capstone (OCSE5707 / OCSE5708) · Open-source release: github.com/mburns00/beyond-ctfs

- Designed a **12-dimensional evaluation framework** (8 performance + 4 safety dimensions) grounded in PTES and MITRE ATT&CK; explicitly addresses gaps in binary CTF-style LLM-pentesting benchmarks
- Conducted a **quasi-experimental, Latin-square repeated-measures comparison** of four **Qwen3-14B** variants (Baseline, SFT, DPO, GRPO) across **N=32 cells** (8 paired HTB machines × 4 variants); applied **triple-blind LLM-as-judge** scoring with variant-redacted transcripts
- Built an **automated evaluation harness** wrapping 10 MCP servers (574 tools) — including pentest, sqlmap, hashcat, cyberchef, wireshark and Playwright tooling — with locked inference config across all runs
- **Statistical pipeline:** Friedman ω -omnibus → Wilcoxon signed-rank (Bonferroni $\alpha = .00833$) → Cliff's δ → Mann–Whitney U → Kruskal–Wallis → Holm–Bonferroni. All 8 performance dimensions significant (χ^2 9.00–15.51, p .029–.0014); 9 of 48 pairwise contrasts Bonferroni-significant
- **Safety evaluation:** N=50 malicious-prompt pack derived from JailbreakBench methodology; mapped observed model behaviour to **OWASP Agentic Top 10, NIST AI 600-1 (12 risks) and Singapore Agentic AI Framework**
- **Inter-rater reliability** reported honestly at **Krippendorff's $\alpha = 0.72$ [0.53, 0.85]** — below the pre-registered ≥ 0.80 target, retained in the report as a null-tolerant finding rather than inflated
- **Key empirical findings:** Pareto-optimal set {Baseline, SFT, GRPO}; DPO weakly dominated by SFT on capability while matching on refusal rate; GRPO exhibits an LLM06 Excessive-Agency signature (96% refusal vs base 90% on malicious pack); detected a DPO training-data contamination fingerprint via behavioural divergence
- **Targeted contributions:** open-source framework, formalised LLM-pentesting failure taxonomy, empirical analysis of SFT/DPO/GRPO trade-offs in offensive-security alignment

Honey Encryption Password Vault with Authenticated Audit Logging · Mar 2026

Applied Cryptography Term Project · Grade: 100/100 · Python / Jupyter

- Implemented Honey Encryption (Juels & Ristenpart, EUROCRYPT 2014) — a Distribution-Transforming Encoder producing plausible decoy plaintexts under brute-force attack; eliminates the authentication-tag leak present in AES-GCM vaults
 - Full cryptographic stack: **PBKDF2-HMAC-SHA256** key derivation, **PRF-XOR** encryption, commitment-based password verification, and **Encrypt-then-MAC (AES-CTR + HMAC-SHA256)** tamper-evident audit logging
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TECHNICAL SKILLS

Offensive Security — Penetration testing (PTES), MITRE ATT&CK, Hack The Box, vulnerability analysis, exploit reasoning, network recon, OWASP Top 10, Linux hardening, digital forensics

AI / LLM Security — Adversarial testing, prompt injection, jailbreak research, model extraction, data poisoning, agentic attack surfaces, OWASP LLM Top 10, MITRE ATLAS

Model Development & Evaluation — LLM post-training (SFT, DPO, GRPO), PyTorch, transformer architectures, evaluation tooling, behavioural testing, failure-mode analysis, LLM-as-judge scoring

Cryptography — AES-GCM/CTR, HMAC, PBKDF2, authenticated encryption, Honey Encryption, Encrypt-then-MAC, formal security definitions (IND-CPA, IND-CCA2)

AI Governance & Risk — NIST AI RMF, NIST AI 600-1, OWASP Agentic Top 10, Singapore Agentic AI Framework, ISO 42001 (familiar), privacy-preserving ML, differential privacy

Programming & Infrastructure — Python, Jupyter, Bash, Git, Linux/WSL, Ollama, automated testing, scientific computing, MCP server development

Cloud Security — AWS (SAA-C03 in progress), Cloud Security specialisation coursework, IT infrastructure, threat modelling

PROFESSIONAL EXPERIENCE

IT Support — Nilrust, Gold Coast QLD · Jan 2023 – Jun 2023

- Delivered hardware, software and network technical support for a manufacturing-sector client base
- Managed and maintained IT infrastructure with a focus on security and operational efficiency
- Trained end-users on internal IT systems and security practices

Technical Support Analyst — Nexigen Digital, Southport QLD · 2020 – 2022

- Diagnosed and resolved escalated technical issues across hardware, software and network layers for a small-to-medium business client base
- Maintained client documentation and ticketing workflows; contributed to repeatable resolution playbooks that reduced first-response handling time
- Supported deployments, system maintenance and incident response across diverse client environments

Technical Support Specialist — TravelSIM, Gold Coast QLD · Dec 2019 – Apr 2020

- Resolved complex customer-facing technical issues in a fast-paced contact-centre environment
- Reduced customer complaints by **75%** through process and triage improvements

Administrative Assistant — Industrial Relations Claims, Brisbane QLD · Aug 2019 – Dec 2019

- Handled sensitive customer information across hardware and software touch-points; resolved cross-system issues spanning IT, scheduling and records management
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RELEVANT COURSEWORK

Applied Cryptography · Cybersecurity Capstone Project A & B · Applied Cybersecurity · Machine Learning & Data Mining · Data Privacy: Theory & Practice · Cybersecurity Engineering · Information Security Management · Professional Practice in IT · Cloud Security · Cloud Computing

ADDITIONAL

- **Australian citizen** — eligible for security clearance (NV1 / NV2 / PV pathways)
- **Open-source:** *Beyond CTFs* evaluation framework (link above)
- **Volunteer:** Manasprings Charity Ball, 2021 – 2023 (venue setup, leadership support)